DSA0503 - CO1 - Assessment Tool 1 - Data Interpretation

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Sales Trend Analysis

A CSV file contains Date, Product, Quantity, and Sales Amount.

Task:

Calculate daily sales.

Find the day with maximum sales.

Plot monthly sales using Matplotlib.

Code:

```python
# Import required libraries
import pandas as pd
import matplotlib.pyplot as plt

# Load CSV file
df = pd.read_csv("sales_data.csv")

# Convert Date column into datetime format
df["Date"] = pd.to_datetime(df["Date"])

# -----------------------------
# 1. Calculate Daily Sales
# -----------------------------
daily_sales = df.groupby("Date")["Sales Amount"].sum()

print("Daily Sales")
print(daily_sales)

# -----------------------------
# 2. Find the Day with Maximum Sales
# -----------------------------
max_day = daily_sales.idxmax()
max_sales = daily_sales.max()

print("\nDay with Maximum Sales")
print("Date :", max_day.date())
print("Sales Amount :", max_sales)

# -----------------------------
# 3. Plot Monthly Sales
# -----------------------------
```

```
monthly_sales = df.groupby(df["Date"].dt.to_period("M"))["Sales Amount"].sum()

plt.figure(figsize=(8,5))
monthly_sales.plot(kind='bar')

plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales Amount")
plt.xticks(rotation=0)
plt.grid(axis='y')

plt.show()
```

Output:

```
Daily Sales
Date
2026-07-01    105000
2026-07-02     52500
2026-07-03     57500
2026-07-04     30000
2026-07-05     60000
2026-07-06     12000
2026-07-07    150000
2026-07-08     15000
2026-07-09     10000
2026-07-10     30000
2026-08-01    100000
2026-08-03      9000
2026-08-05     45000
2026-08-07     75000
2026-08-09      9000
Name: Sales Amount, dtype: int64

Day with Maximum Sales
Date : 2026-07-07
Sales Amount : 150000
```